

Characters of Representations, Briefly

Jonas Carew

July 26, 2026

Contents

Representations and Fundamental Results 2

Class Functions and Characters 7

The Character Correspondence 11

Character Tables 14

Big Picture 18

Representations and Fundamental Results

Representations and Fundamental Results

What is a Representation?

Given a group G and a vector space V , a *representation* of G over V is a group homomorphism

$$\rho : G \rightarrow \mathrm{GL}(V)$$

We focus now on the usual case of G being a finite group and V being a $\mathbb{C}$ -vector space, although many results hold in further generality.

Why do we care?

Representations allow us to study groups by studying matrices. In particular, a *faithful* representation is one where ρ is injective, giving us an isomorphic copy of G inside of a matrix group.

Representations and Fundamental Results

Examples

The *trivial representation* ρ of G on a one dimensional space T is $\rho(g) = I$ for all $g \in G$.

For $G = \mathbb{Z}/3\mathbb{Z}$, we get a representation $\rho : G \rightarrow \mathbb{C}^\times$ by

$$\rho(0) = 1 \quad \rho(1) = \omega \quad \rho(2) = \omega^2$$

where ω is the primitive third root of unity, $\omega = e^{\frac{2\pi i}{3}}$. We can think of these as rotations of the complex plane; there is a similar representation for any cyclic group.

If we let $\mathbb{C}G$ be a $\mathbb{C}$ -vector space with basis indexed by the elements of G , we can define the *left regular representation* $\lambda : G \rightarrow \text{GL}(\mathbb{C}G)$ by

$$[\lambda(g)](\alpha_1 e_{g_1} + \cdots + \alpha_n e_{g_n}) = \alpha_1 (e_{gg_1}) + \cdots + \alpha_n (e_{gg_n})$$

where $n = |G|$. We call $\mathbb{C}G$ the *group algebra* of G .

Representations and Fundamental Results

Convention

We notice that a representation $\rho : G \rightarrow \mathrm{GL}(V)$ gives us an action of G onto V by

$$g \cdot v = [\rho(g)](v)$$

and we adopt this notation from here on.

Building New Representations

Given representations V and W of G , we can form new representations:

- G acts on $V \oplus W$ by $g \cdot (v, w) = (g \cdot v, g \cdot w)$
- G acts on $V \otimes W$ by $g \cdot (v \otimes w) = (g \cdot v) \otimes (g \cdot w)$

Maschke's Theorem

Given a subrepresentation W of V , it has a complementary subrepresentation W' such that $V = W \oplus W'$. In this way, representations of finite groups are indecomposable exactly when they are irreducible, which is not true in general.

Representations and Fundamental Results

Factoring Representations

These results lead to a major result in the representation theory of finite groups:

For any representation V of G we can write

$$V \simeq V_1^{\oplus n_1} \oplus \cdots \oplus V_k^{\oplus n_k}$$

where each V_i is a distinct irreducible representation of G .

This means that studying the irreducible representations of G tells us how *every* representation of G acts.

Class Functions and Characters

Class Functions and Characters

Class Function

A *class function* is a function $\alpha : G \rightarrow \mathbb{C}$ that is constant on conjugacy classes, i.e., $\forall g, h \in G$

$$\alpha(g) = \alpha(hgh^{-1})$$

The class functions on a group form a $\mathbb{C}$ -vector space.

Inner Product

We can take the inner product of two class functions α and β by

$$\langle \alpha, \beta \rangle = \frac{1}{|G|} \sum_{g \in G} \overline{\alpha(g)} \beta(g)$$

Under this inner product, the indicator functions for the conjugacy classes form an orthogonal basis for the space of class functions.

Class Functions and Characters

Characters

The *character* $\chi_V : G \rightarrow \mathbb{C}$ of a representation $\rho : G \rightarrow \mathrm{GL}(V)$ is defined as

$$\chi_V(g) := \mathrm{tr}(\rho(g))$$

Using Characters

Since characters use the trace which is the sum of the eigenvalues, we get that

$$\chi_{V \oplus W} = \chi_V + \chi_W \quad \chi_{V \otimes W} = \chi_V \chi_W \quad \chi_{V^*} = \overline{\chi_V}$$

and since $\rho(e) = I$, we get that $\chi_V(e) = \dim(V)$.

Class Functions and Characters

Characters as Class Functions

Since the trace of a matrix is a group homomorphism from $\mathrm{GL}(V) \rightarrow \mathbb{C}^\times$ and multiplication in $\mathbb{C}^\times$ is commutative, we see that

$$\chi(hgh^{-1}) = \mathrm{tr}(\rho(hgh^{-1})) = \chi(h)\chi(g)\chi(h)^{-1} = \chi(g)$$

Irreducibles form a Basis

The characters for the irreducible representations of G form an orthonormal basis for the space of class functions. This motivates us to study the distinct irreducible characters of G .

The Character Correspondence

The Character Correspondence

Corollaries

Because the irreducible characters form a basis for the rest of the characters, we get several very important corollaries:

- Representations V and W are equivalent if and only if $\chi_V = \chi_W$.
- A representation V is irreducible if and only if $\langle \chi_V, \chi_V \rangle = 1$.
- If V is an irreducible representation, the multiplicity of V in the decomposition of W is obtained by $\langle \chi_V, \chi_W \rangle$.

Since the indicators functions for the conjugacy classes of G also form a basis for the space of class functions, the number of distinct irreducibles is in bijection with the number of conjugacy class of G , but there is no natural correspondence in general.

The Character Correspondence

Properties of The Regular Representation

Recall $\lambda : G \rightarrow \text{GL}(\mathbb{C}G)$ acts by $g \cdot e_h = e_{gh}$. This means that $\lambda(g)$ is a permutation matrix and the character χ is the number of fixed points, thus giving us

$$\chi(g) = \begin{cases} |G| & \text{if } g = e \\ 0 & \text{if } g \neq e \end{cases}$$

We then see that for an irreducible representation V ,

$$\langle \chi_V, \chi \rangle = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \chi(g) = \overline{\chi_V(e)} = \dim(V)$$

Thus the decomposition of the regular representation gives two very important identities:

$$\mathbb{C}G \simeq V_1^{\oplus \dim(V_1)} \oplus \cdots \oplus V_k^{\oplus \dim(V_k)} \quad \text{and} \quad |G| = \chi(e) = \dim(\mathbb{C}G) = \sum (\dim V_i)^2$$

Character Tables

Character Tables

Column Invariance

This last result can be generalized, giving us the formula

$$\frac{1}{|G|} \sum \overline{\chi_{V_i}(g)} \chi_{V_i}(h) = \begin{cases} 1/c(g) & \text{if } g \text{ and } h \text{ are conjugate} \\ 0 & \text{otherwise} \end{cases}$$

where $c(g)$ is the order of the conjugacy class of g . In particular, this gives us

$$|G| = c(g) \sum |\chi_{V_i}(g)|^2$$

which can be seen visually in a character table.

Character Tables

Character Table

The *character table* for a group is a square table listing all of the conjugacy classes, irreducible representations, and the values of each character on each class.

Example

For $G = \mathbb{Z}/3\mathbb{Z}$, we see that since the group is abelian, each element has its own conjugacy class.

The two nontrivial representations are obtained by sending 1 to the primitive 3rd roots of unity.

Since the conjugacy classes are in bijection with the irreducible representations, the table is square.

	1	1	1
$\mathbb{Z}/3\mathbb{Z}$	[0]	[1]	[2]
T	1	1	1
V_2	1	ω	ω^2
V_3	1	ω^2	ω

Character Tables

Example

For $G = A_4$, we take $N = \langle (12)(34), (14)(23) \rangle$ and see that $A_4/N \simeq \mathbb{Z}/3\mathbb{Z}$. Thus we can extend our previous representations of $\mathbb{Z}/3\mathbb{Z}$ to A_4 as such:

$$\rho : G \rightarrow G/N \rightarrow \mathrm{GL}(V)$$

We see that the first column then allows us to solve for the dimension of V_4 using

$$|A_4| = \sum (\dim V_i)^2 = \sum \chi_i(e)^2$$

and the column invariance further gives us

$$|A_4| = c(\sigma) \sum |\chi_i(\sigma)|^2$$

	1	4	4	3
A_4	(1)	(123)	(132)	(12)(34)
T	1	1	1	1
V_2	1	ω	ω^2	1
V_3	1	ω^2	ω	1
V_4	3	0	0	-1

Big Picture

Questions?

Final Overview

- Use *group representations* to study finite groups as matrices.
- Since all representations are fully decomposable, we only study the *irreducibles*.
- The irreducible *characters* form an orthonormal basis for class functions, giving us many nice properties.
- We can collect all of this information into a *character table*, allowing us to see the relationships easier.

Questions?